Course Code: SECI1143-03

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Project 2 : Inferential Statistical Analysis

Group Name: False 9

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1.0 : Introduction

Academic success in core subjects like mathematics is a major stepping stone for a student's future career and higher education prospects. However, a student's final grade is rarely just a reflection of their intelligence.This raises an important question: what really drives high school performance?To uncover these answers, this project investigates the relationships between students' personal lives and their mathematics grades using statistical methods like hypothesis testing, correlation, regression analysis, and the chi-square test of independence.To conduct this analysis, we are utilizing a secondary dataset obtained from Kaggle. The data was originally gathered by researchers P. Cortez and A. Silva for a 2008 study focusing on secondary school student performance in Portugal.

- To practically apply inferential statistical tests to a real-world secondary dataset.
- To mathematically determine whether significant relationships and dependencies exist between a student's external environment and their final academic performance.

2.0 : Dataset

2.1 Description of Data

Population :
Sample: 395 students
Data Description:

Variables (Description)	**Type of Variable**	**Measurement Level**
sex (student gender)	Qualitative	Nominal
study_time (weekly study time)	Qualitative	Ordinal
higher_ed (desire for higher education)	Qualitative	Nominal
internet_access (internet access at home)	Qualitative	Nominal
absences (number of school absences)	Quantitative	Ratio
final_grade (final grade)	Quantitative	Ratio

2.2 Statistical Test Analysis

Refer to Appendix 1A

3.0 : Data Analysis

3.1 : Two Sample Hypothesis Test

3.1.1 : Variables and Objective

In this analysis,we use the variables of final_grade and sex.The purpose of this study is to test whether there is a significant difference in the average final grade of Mathematics between female students (F) and male students (M).

3.1.2 : Hypothesis Statement

Null Hypothesis (H_0) : There is no significant difference in the average final score between female and male students.

Alternative Hypothesis (H_1) : There is a significant difference in the average final_grade between female and male students.

$H_0 : \mu 1 = \mu 2$

$H_1 : \mu 1 \neq \mu 2$

($\mu 1$ represent average mark of female students,while $\mu 2$ represent average mark of male student)

3.1.3 : Interpretation Of The Test

Based on a Two-Sample t-Test analysis conducted the table below is a summary of the statistical results:

t_0 (test statistic)	-2.0651
df	390.57
p-value	0.03958
Mean Of Female	9.966346
Mean Of Male	10.914439

The test was conducted at a significance level $\alpha = 0.05$.

Therefore,we will reject H_0 if p-value < 0.05.

Since the p-value (0.03958) is smaller than the significance level (0.05),we **reject** the null hypothesis.Hence There is sufficient evidence at the 5% significance level to state that there is a significant difference in the average final score between female and male students.

3.1.4 : Conclusion

In conclusion,by looking at the group means, male students achieved a higher average final grade (10.914) compared to female students (9.966). This indicates that male students performed slightly better in Mathematics within this sample. This insight could be further investigated to explore whether external factors such as learning styles, background, or interest levels contribute to this performance gap between genders.

3.2 : Correlation Analysis

In this analysis, we will use the variables study_time and final_gradel to determine whether there is a linear relationship between study time and final grade levels at a 95% confidence level using Pearson's Product-Moment Correlation Coefficient. The strength of association, which is the linear relationship between two variables, is measured using correlation analysis. We use Pearson's Product-Moment Correlation Coefficient for the correlation coefficient because the variables study_time and final_gradel are both ratio -type data.

Scatter Plot of Study Time vs Final Grade (Refer to Appendix A)

According to the scatter plot above, there will be a weaker positive correlation relationship between the study time and final grade. The relationship is the fact that the more study time, the higher your final grade will be. However, there are a few outliers above the plot as well.

By using RStudio, we can see that the sample correlation coefficient, r is 0.0978, which indicates that there is a very weak positive linear correlation between the study time and final grade.

2. Significance Test for Correlation

- Hypothesis Statement:

  H_0: $\rho = 0$ (no linear correlation)

  H_1: $\rho \neq 0$ (linear correlation exists)

By using RStudio, we get the test statistic, t is 1.9485.

● Find critical value, using α = 0.05, df = n-2 = 393

From t-table, since this is a two-tailed test, there are two critical values:

Lower tail critical value : t(0.025,393) = -1.966
Upper tail critical value : t(0.075,393) = 1.966

From RStudio, we also get p-value = 0.05206.

Hence, if test statistic > 1.966 or test statistic < -1.966, we will reject null hypothesis, H_0 . Otherwise, we will fail to reject the null hypothesis, H_0 .

● State the decision:

Bell curve graph

Refer to Appendix B

Since the test statistic, t = 1.9485 < upper tail critical value which is t(0.075,393) = 1.966, we fail to reject the null hypothesis. In conclusion, there is no sufficient evidence to conclude that there is a linear relationship between the study time and final grade of students, at α = 0.05.

Conclusion and Discussion of the Analysis

The correlation analysis reveals a positive but weak relationship between a student's weekly study time (study_time) and their final academic performance (final_grade). Because study time is measured in ordered categories rather than exact hours, the modest correlation indicates that while studying longer generally trends toward higher marks, it is not a standalone guarantee of success. The data shows significant variance: many students achieve high grades with minimal study hours, while others log maximum study time but continue to face academic difficulties. Thus, while statistically meaningful, study time alone plays a relatively minor role in predicting a student's absolute final score.

3.3 : Regression Analysis

Variables used

In this simple linear regression analysis, the independent variable (x) is the number of school absences (absences) a student has accumulated throughout the academic year. The dependent variable (y) is the student's final mathematics grade (final_grade), measured on a scale from 0 to 20.

Objective of the Test

The primary objective of this regression test is to determine whether a statistically significant linear relationship exists between a student's attendance record and their overall academic success. Specifically, this analysis aims to predict how much a student's final mathematics grade is expected to decrease or increase for every additional day of school they miss.

Scatter Plot : Absences vs Final Grade (Refer to Appendix C)

Results and Interpretation

According to the RStudio results,the test statistic are :

b_0 (Intercept) = 10.30327 ,b_1 (Slope) = 0.01961.

sb_1 (Standard Error of the Slope) = 0.02886. $s\varepsilon$ (Residual Standard Error) = 4.585.

df (Degrees of Freedom) = 393. R^2 (Multiple R-squared) = 0.001173. p-value = 0.4973.

Hypothesis Statement :

Null Hypothesis (H_0) : There is no statistically significant relationship between the number of school absences and a student's final_grade of mathematics

Alternative Hypothesis (H_1) : There is a statistically significant relationship between the number of school absences and a student's final_grade of mathematics

H_0: $\beta_1 = 0$ (no linear relationship)

H_1: $\beta_1 \neq 0$ (linear relationship does exist)

Find critical value, using $\alpha = 0.05$, df = n - 2 = 393: Since this is a two-tailed test, the two critical values gotten from RStudio are:

$t_{0.025,\ 393} = 1.966$

$-t_{0.025,\ 393} = -1.966$

Hence, we reject H_0 if the test statistics < -1.966 or test statistics > 1.966.

Calculate test statistics using formula: By using RStudio, we get the test statistic $t_0 = 0.679$

State the decision:

Bell curve graph (**Refer to Appendix D)**

Since test statistics $t_0 = 0.679 < t_{0.025,\ 393} = 1.966$, we fail to reject the null hypothesis. There is insufficient evidence to prove that there is a statistically significant relationship between the number of school absences and a student's final_grade of mathematics at $\alpha = 0.05$.

Conclusion and Discussion of the Analysis

Based on the simple linear regression model and the subsequent t-test conducted, we conclude that there is no statistically significant linear relationship between the number of school absences a student accumulates and their final mathematics grade. In a real-world context, this is an insightful finding because it challenges the standard assumption that missing school automatically leads to lower grades. This result suggests that students who miss classes might be successfully compensating for their absence through other means. -Finally, we can add the linear regression model into the scatter plot:

Regression Line : Absences vs Final Grade (Refer to Appendix E)

Based on the scatter plot above, the data points are widely dispersed across the entire graph, failing to cluster closely around the regression line. This wide scattering visually confirms our extremely low Multiple R-squared value of 0.00117, illustrating that the regression model is a very poor fit for predicting the data.The regression line itself is nearly horizontal, displaying only a microscopic positive slope ($b_1 = 0.020$).

3.4 : Chi-Square Test of Independence

In this analysis, we will use the variables **internet_access** and **higher_ed** to assess the significant relationship between the independence of home internet access and higher education aspirations using a Two Way Contingency Table at a 95% confidence level. As a result, we employ the Chi-Square Test of Independence in conjunction with a two-way contingency table **.**

Observed Frequencies for variables **internet_access** and **higher_ed**

3.4.1 State the test hypothesis :

H_0: Both of the variables are independent

H_1: Both of the variables are dependent

1. Find the critical value:

   Critical value, $x^2 = 3.841$

   (with df=(2-1)(2-1)=1, =0.05)

By using RStudio, the value of the critical value , x^2 is 3.841459.

Refer to Table (Refer in Appendix F)

When we calculate test statistic manually, we get test statistic, $x^2 = 0.1649$

Using R Studios,

When we calculate test statistic using RStudio, we also get test statistic, $x^2 =0.16396$, with the p-value = 0.6855.

Refer to Table

3.4.2 State the decision

Bell curve graph (Refer to Appendix G)

Since the test statistic value ($x^2 =0.16396$) < critical value ($x^2_{df=1,\alpha=0.05} = 3.841$), it does not fall within the critical region. Thus, we fail to reject the null hypothesis, H0. Therefore, there is not sufficient evidence to conclude that there is a relationship between the variables internet_access and higher_ed at $\alpha = 0.05$.

3.4.3 Conclusion of Chi-Square

In conclusion, by looking at the categorical frequencies, the vast majority of students aspire to pursue higher education regardless of whether they have access to home internet or not. This indicates that a student's academic ambitions are not limited by home internet availability within this sample. This insight could be further investigated to explore whether external factors such as school resources, parental encouragement, or personal career goals contribute more decisively to a student's motivation for higher education than technology access at home.

4.0 : Conclusion

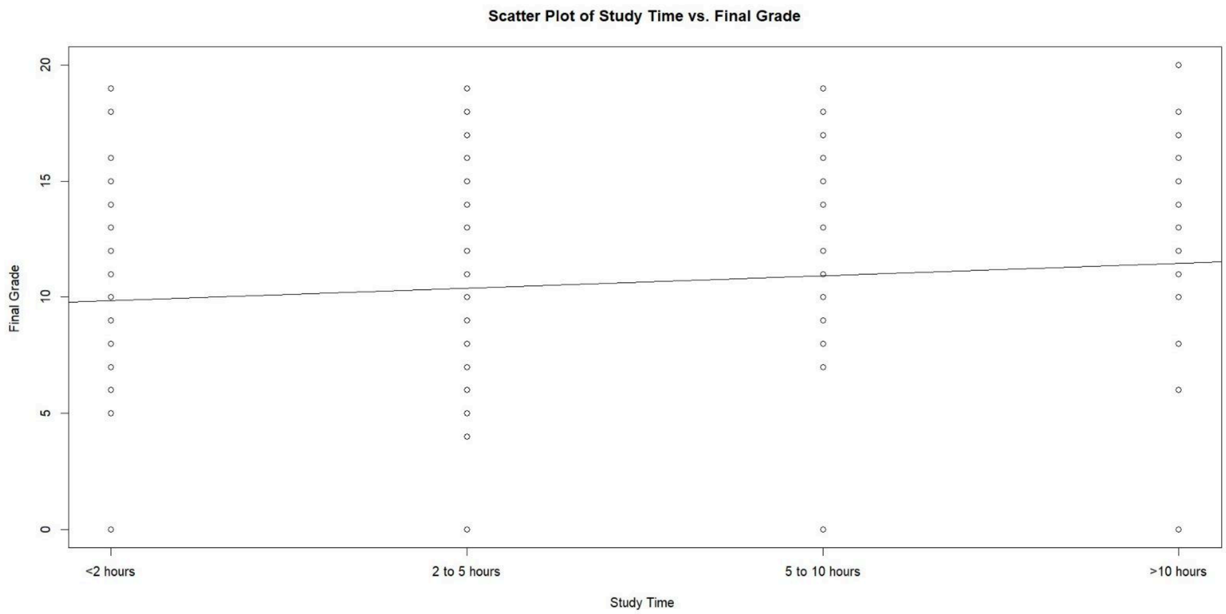
Through the implementation of Project 2, we have practically learned the complete data science cycle using RStudio starting from the dataset selection phase, data pre-processing process, to analysis techniques. Among the most interesting and surprising findings from our study results is that quantitative and logistical factors do not determine a person's academic success; although Mathematics grades show differences according to gender (male students get higher averages), the performance is not affected at all by the number of weekly study hours or school absence records, while aspirations to further their studies have proven to be independent of the influence of internet access at home.

5.0 : Appendix
Appendix 1A

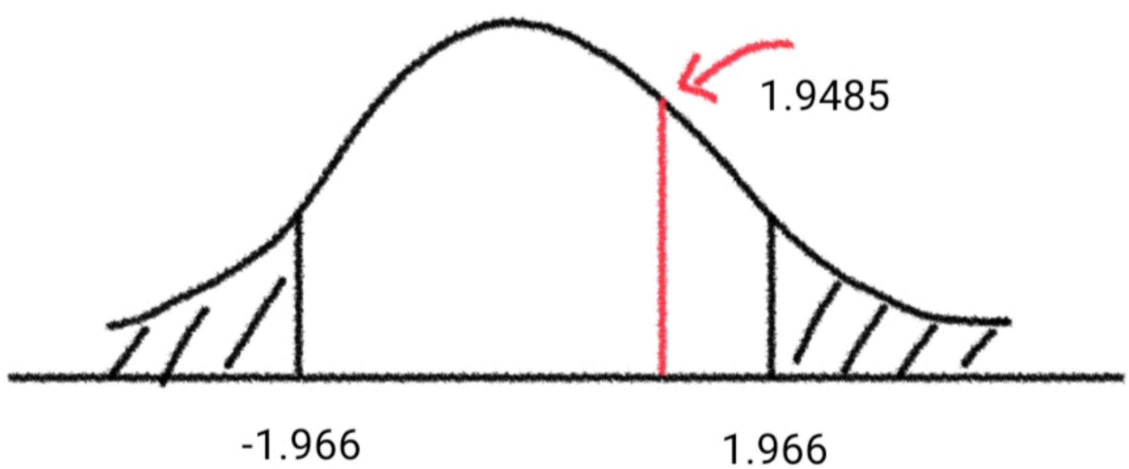
Selected Variables	Objectives	Test Analysis and Expected Outcome
sex, final_grade	To determine whether the final mathematics grade differs between male and female students.	**Analysis:** Independent Two-Sample t-Test (Variance Unknown) **Expected Outcome:** By conducting a two-sample t-test, we can compare the mean final grade between male and female students. If the p-value is less than 0.05, it suggests a significant difference in academic performance between the two groups.
study_time, final_grade	To examine the relationship between study time and final mathematics grade.	**Analysis:** Correlation Analysis (Pearson Correlation) **Expected Outcome:** By calculating the correlation coefficient, we can determine the strength and direction of the relationship between study time and final grade. A positive and significant correlation

		would indicate that students who study longer tend to achieve higher grades.
absences, final_grade	To assess whether the number of absences predicts students' final mathematics grade.	**Analysis:** Simple Linear Regression **Expected Outcome:** By performing simple linear regression, we can determine whether absences significantly affect final grade. A significant negative regression coefficient would suggest that increased absences lead to lower academic performance.
internet_access, higher_ed	To evaluate the independence between internet access at home and students' desire to pursue higher education.	**Analysis:** Chi-Square Test of Independence **Expected Outcome:** By conducting a chi-square test, we can determine whether internet access and higher education aspirations are associated. If the p-value is less than 0.05, it suggests a significant relationship between the two variables.

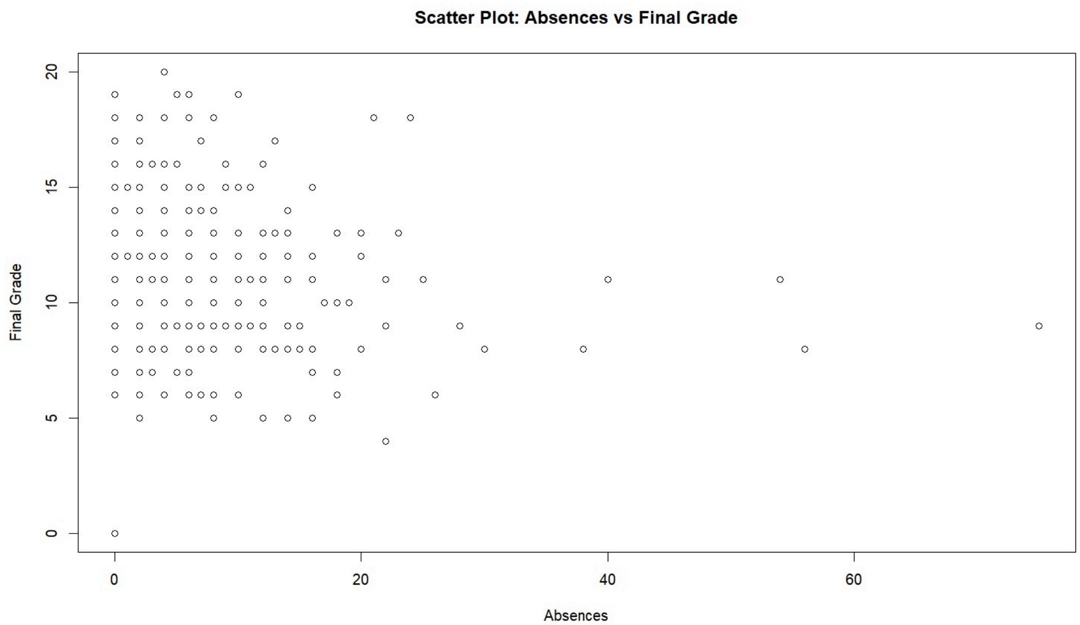
Appendix A



Appendix B

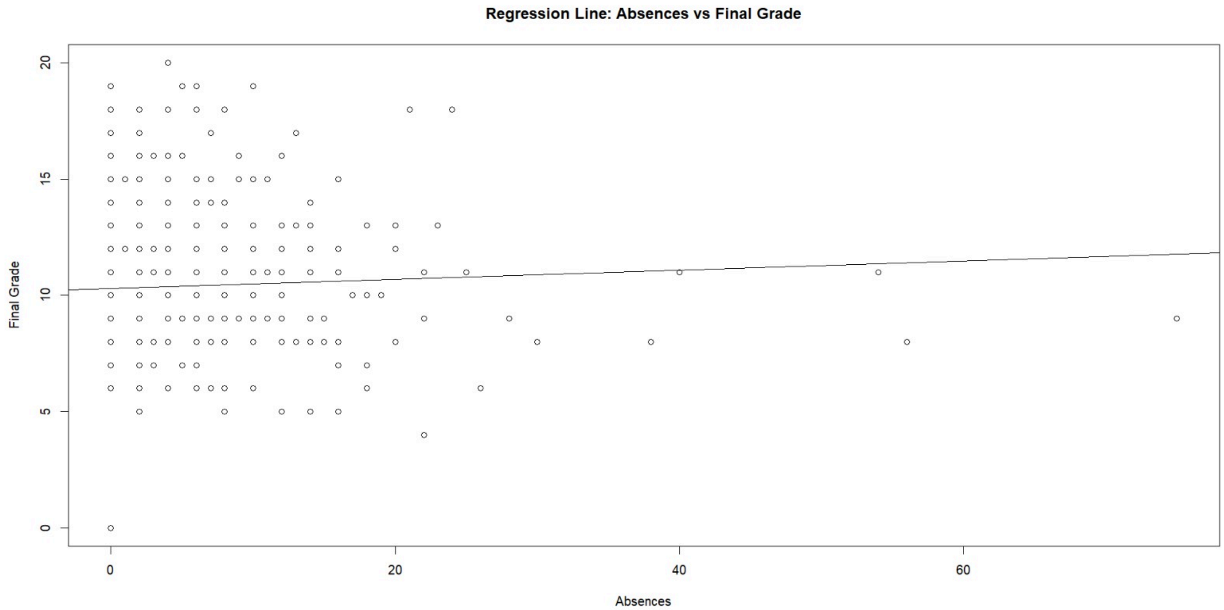


Appendix C



Appendix D

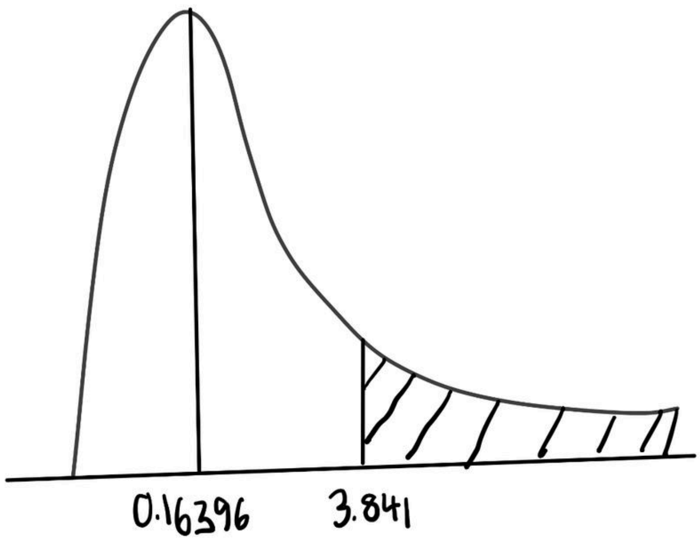
Appendix E



Appendix F

internet_access	higher_ed				Total
	0		1		
	Obs.	Exp	Obs.	Exp	
0 (No)	4	$\frac{66x20}{395} = 3.34$	62	$\frac{66 \times 375}{395} = 62.66$	66
1 (Yes)	16	$\frac{329x20}{395} =16.66$	313	$\frac{329x375}{395} = 312.34$	329
Total	20	20	375	375	395

Appendix G



Video Link Presentation : https://youtu.be/Mb-3e2Vm4hQ?si=D-M2K1tS3rjHFH6D